

65. (d) **Nitrogen** does not contribute to the greenhouse effect.

Greenhouse effect is the phenomenon by which certain gases present in the atmosphere trap heat i.e do not allow the heat to escape the Earth.

CO₂ is one of the greenhouse gases. An increase in the amount of carbon dioxide in the atmosphere can cause more heat to be retained and hence leads to global warming.

66. (c) Prairies - USA

Downs - Australia

Llanos - Venezuela

Pampas - South America

67. (b) While methane, oxides of sulfur and nitrogen, and carbon dioxide are all greenhouse gases that contribute to global warming by trapping heat in the Earth's atmosphere, **water vapor** is not considered a significant contributor in terms of human-induced climate change.

Although water vapor is a natural greenhouse gas and plays a crucial role in the Earth's climate system, its concentration in the atmosphere is primarily controlled by temperature and natural processes.

68. (d) **Operation Nistar** was a rescue operation conducted by the Indian government to evacuate Indian nationals from Yemen during a crisis. The operation took place in 2015 and involved the evacuation of Indian citizens from conflict-affected areas in Yemen.

Operation Pawan was a military operation conducted by the Indian Peace Keeping Force (IPKF) in Sri Lanka. It aimed at controlling the Jaffna peninsula during the Sri Lankan Civil War.

Operation Trident was a naval operation conducted by the Indian Navy during the Indo-Pakistani War of 1971. It was a significant and successful naval offensive carried out on the night of December 4-5, 1971, against Pakistan.

Operation Brasstacks was a code name for a military exercise conducted by the Indian Armed Forces in 1986. It was one of the largest military exercises conducted by India.

69. (b) The **PRITHVI** missile is a surface-to-surface tactical short-range ballistic missile developed by India's Defence Research and Development Organisation (DRDO).

It is named after the Prithvi, the Earth in Sanskrit, and has multiple variants, including Prithvi-I, Prithvi-II, and Prithvi-III, each with different ranges and capabilities.

70. (d) The launch of India's first Flood Forecasting and Early Warning System

(FFEWS) for Kolkata City is a significant step towards enhancing the city's resilience to flooding and improving disaster preparedness.

The Flood Forecasting and Early Warning System (FFEWS) for Kolkata City has been launched by the Mayor of Kolkata Municipal Corporation (KMC).

71. (b) Under the chairmanship of Prime Minister Narendra Modi, the Prime Minister's Citizen Assistance and Relief in Emergency Situation Fund was created on March 28th, 2020.

The Prime Minister's Citizen Assistance and Relief in Emergency Situations Fund popularly known as PM-CARES Fund has been established by the government to address any emergency or distress scenario, including the COVID-19 pandemic.

It gives facility to avail donation from the foreign contribution and donation to fund can also avail 100% tax exemption.

72. (d) All three factors contribute to earthquakes:

Location: The tectonic plate boundaries play a crucial role in determining the occurrence of earthquakes.

Distance from the epicenter: The epicenter is the point on the Earth's surface directly above the earthquake's point of origin (hypocenter).

Local geological conditions: The geological characteristics of an area can influence how seismic waves travel through the Earth.

73. (c) Earthquakes are primarily caused by the movement of tectonic plates beneath the Earth's surface, leading to the release of energy in the form of seismic waves.

Distance from the epicenter, location, and local geological conditions are significant factors influencing the occurrence and intensity of earthquakes.

These factors include proximity to fault lines, the type of rock and soil, and the presence of geological features that can amplify seismic waves.

74. (c) Assertion (A) correctly states that the Portuguese were the first European community to discover a direct sea route to India, which they achieved in 1498 under the leadership of Vasco da Gama.

Reason (R) accurately explains the context behind this discovery. The levying of excessive taxes by the Turks on goods passing through traditional trade routes between Asia and Europe incentivized European powers like Portugal to seek alternate sea routes to access valuable goods from the East, such as spices and textiles, directly.

75. (c) According to Newton's third law, for every action, there is an equal and opposite reaction.

When a person jumps out of a boat, they exert a downward force on the boat due to their push-off. Simultaneously, the boat exerts an equal and opposite force upward on the person, as per Newton's third law.

This action-reaction pair of forces causes the boat to move in the opposite direction to the person jumping out, ensuring that the total momentum of the system (person + boat) remains conserved.

76. (b) 

Some Furnitures are made from Maple's treewood

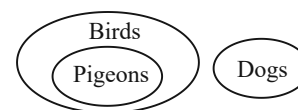
Some Furnitures are made from Oak's treewood

77. (a) The logic followed here is:

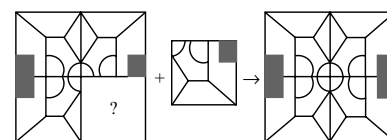
The movement of Rabbits is Hop.

Similarly, the movement of Duck is known as Waddle.

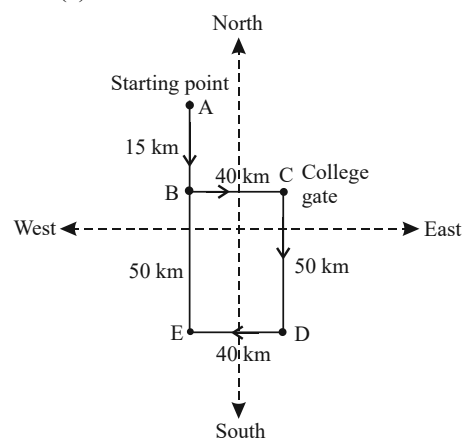
78. (d) Pigeons are a specific type of birds. Thus pigeons will be completely inside birds. And Birds and Dogs are mutually exclusive of each other that mean they will not intersect each other.



79. (d)



80. (a)



Minimum distance between AE

$$= 50 + 15$$

$$= 65 \text{ km}$$

